

The Invisible Data: Mapping the Perimenopausal Data Gap in Women's Health

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Abstract

Perimenopause is a midlife women's health issue which impacts millions of women worldwide, yet it remains underdiagnosed and poorly represented in clinical research and health technologies. This study examines how perimenopausal experiences are articulated, interpreted, and supported within digital communities by utilizing user-generated discussions from the Reddit forum 'r/perimenopause'. Grounding this research in human centered design methods, feminist HCI principles, and narrative data visualization techniques, helps ensure that lived experiences of women, which are deeply personal and historically underrepresented are centered rather than marginalized within digital health technologies. Together, these approaches address issues of equity and representation while translating complex qualitative experiences into empathetic, meaningful design solutions. Using an iterative inductive coding process, my research identifies underlying themes and patterns related to physical, psychological, and emotional symptoms; information seeking behaviors; coping mechanisms; and experiences of validation and frustration within healthcare systems. A summative content analysis was conducted on 1000 posts and 39,698 comments dated between August 2023 and June 2025, providing a large-scale qualitative dataset for thematic analysis. Findings show the wide range of perimenopausal symptoms and underscore the role of online forums in helping women make sense of their experiences together. To explore complementary methods of analyzing large corpuses of qualitative data, a GPT-4 classifier was trained on hand annotated data to analyze symptoms with results indicating the need for further evaluation of this approach. Finally, these insights are disseminated through narrative driven visualizations designed to support both analytical interpretation and intuitive understanding. The findings aim to reduce data gaps by translating lived perimenopausal experiences into insights that inform empathetic digital health tools, support symptom management, and advance human-centered approaches in HCI.

CCS Concepts

• **Computing methodologies** → Artificial intelligence; • **Human-centered computing** → HCI theory, concepts and models; *User centered design*; **Information visualization**; **Visual analytics**;

Keywords

Mid-life women's health, lived experience, perimenopause, feminist HCI, digital health communities, social data analysis

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1 Introduction

According to data from the U.S. Census Bureau, about 63% of women in the United States are in midlife; yet health education and research focused on this important and transitional stage in a woman's life have been under-prioritized and underfunded [16]. One such challenging phase is perimenopause. Due to gradual decline of ovarian function and estrogen levels, women face a complex array of clinical symptoms such as irregular menstrual cycles, decline in cognitive health, vasomotor symptoms, palpitations, vaginal dryness, sexual dysfunction, and emotional volatility [14] [13]. Perimenopause can differ in its onset, range of symptoms, pattern, severity, but it is generally noted that it can start from early 30s and last for anywhere between 5-15 years. Perimenopause culminates with menopause, when menses have ceased for a period of at least 12 consecutive months [4]. Despite affecting a significant segment of the population, perimenopause is largely understudied in both clinical and socio technical literature, particularly in the context of design and human-computer interaction (HCI) [12]. Socially, conversations surrounding midlife women's health have often been gaslighted, stigmatized, or silenced, contributing to a lack of awareness and shared knowledge about midlife women's health [2] [9]. Structurally, this stage is poorly defined in clinical settings, it lacks biomarker precision and standardized diagnostic criteria which makes it difficult to study medically. Additionally, physicians' level of knowledge in addressing the concerns vary greatly, limiting the support they provide to women in need [10]. As a result, many women navigate this complex phase without adequate support, care, knowledge, tools, and medical assistance which makes it one of the most misunderstood chapters in a female's reproductive lifespan. Amidst the lack of clinical support, women experiencing perimenopause often take to online forums to commiserate and share experiences from their everyday lives. Online forums and platforms like Reddit, Quora, Facebook groups, and Instagram pages focusing on midlife women's health have created a form of "virtual sisterhood" where women share their experiences, find validation, and exchange information, forming a sense of community



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and solidarity [6]. Discussions about women’s health on Reddit increased by 37% from 2023 to 2024, and a large pool of users reported feeling less isolated in navigating their healthcare by participating in those communities [17]. Therefore, these forums are valuable repositories of qualitative data to supplement a more comprehensive understanding of perimenopausal life.

2 Related Work

A growing number of researchers have begun to reclaim reproductive health and hormonal phases in women’s lives, focusing on reproductive health [11], menstrual stigma [3], and health literacy [7]. In contrast, perimenopause has received far less creative or critical engagement, particularly within HCI. Amanda Lazar et al. [8] conducted a sociotechnical analysis by studying discussions in a subreddit forum to understand menopausal experiences and found that discussions around bodily changes are shaped by gendered marginalization, social relationships, and cultural context. The authors found that women use Reddit to construct meaning around menopause through shared language and storytelling which led to community-based narration and empowerment. Sillence et al. conducted a longitudinal study to assess the effectiveness of the “Health and Her” mobile application in supporting symptom management during perimenopause and menopause and found that higher levels of engagement through features like symptom tracking, period logging, journalling, and participation in self-care activities were significantly associated with reduced symptom severity [15]. Further, they also discussed the potential healthcare apps can have in giving agency to midlife women by providing self-management and positioning these tools as valuable tools in the broader context of FemTech and digital health products. Together, these studies suggest that while digital tools for women’s midlife health are gaining traction and show potential in empowering women; to be successful, these interventions must include women’s lived experiences, and must be user centric, holistic, and credible.

3 RESEARCH APPROACH AND METHODS

Using human centered design methods, summative content analysis methods, narrative data visualization techniques, and feminist HCI principles, this study treats women’s online narratives as legitimate forms of knowledge production rather than merely anecdotal data. This approach centers marginalized voices and seeks to translate experiential insights into empathetic visualizations.

3.1 Research Questions

At large, this research will address the following questions:

- How are symptom management and information exchange addressed in women’s health care?
- How can data from online community forums help identify hidden patterns in perimenopausal symptoms?

3.2 Research Design

3.2.1 Data Corpus: User-generated discussions within the Reddit community ‘r/perimenopause’, a peer supported forum dedicated exclusively on perimenopausal experiences was used for this study. A comprehensive whole person approach has been followed here

where the focus is not only on physical symptoms but also individual lived experiences [1]. Despite the dataset being cleaned after extraction, the qualitative, text-based data may contain interpretive variability [5].

3.2.2 Procedure and Analysis: The dataset spans posts dated between August 2023 and June 2025 and includes 1000 posts and 39,698 comments. Data was collected using the Reddit API in combination with web scraping techniques and was pre-processed using Python based tools like Pandas and Natural Language Toolkit (NLTK).

- (1) **Analysis 1- Language Mapping:** To understand how women speak about symptoms in community forums, all posts and comments were reviewed to become familiar with the dataset. A table was then created to map informal terms used to describe physical, cognitive, emotional, and psychological symptoms. Through this analysis, a distinctive pattern emerged where individuals described symptoms using vivid metaphors such as “zombie mode,” “blood bath,” or “feels like I’m on heavy steroids.” These expressions illustrate how language is used within the community to communicate intense bodily experiences.
- (2) **Analysis 2- Keyword Frequency Analysis:** Using the table developed in Analysis 1, a keyword-based analysis was conducted to calculate the frequency of each symptom. Informal expressions used by women were grouped with their corresponding medical terms, after which symptom frequencies were computed. Experiences such as emotional and psychological factors—often lacking clear diagnostic tests—are frequently overlooked in clinical settings. Capturing and quantifying these experiences helps highlight the significance of this phase and amplifies voices often marginalized in traditional medical discourse.
- (3) **Analysis 3- Symptom Co-occurrence Analysis:** To understand how symptoms appear together, a symptom co-occurrence matrix was created in which each comment was coded for the presence or absence of individual symptoms. Pairwise distances between symptoms were calculated to quantify co-occurrence patterns, and hierarchical clustering was applied to the resulting distance matrix. The dendrogram revealed groups of symptoms that frequently occur together.
- (4) **Analysis 4- Coping Strategy Identification:** To identify coping strategies discussed in the dataset, a combination of keyword searching and manual thematic categorization was used to group comments into strategy categories (e.g., medical treatments, mindfulness practices, and lifestyle-based approaches). Frequencies were calculated to quantify how often each strategy was mentioned, and results were visualized using log-scaled plots to capture the wide range of counts across categories.

3.2.3 Use of AI: A complementary method was explored to understand the utility of artificial intelligence (AI) for analyzing large corpus of noisy qualitative data. Open AI’s Generative Pre-Trained Transformer 4 (GPT) classifier was used to support symptom classification. A subset of 250 comments was used to develop and refine

Table 1: Informal descriptions of perimenopausal experiences discussed in Reddit forums.

Symptom	Category	Informal Phrases
Heavy Period Flow	Physical	Blood bath; period from hell; flooding
Palpitations	Physical	Heart jumping; racing heart; thumping; heart pounding
Weight Gain	Physical	Extra fluff; clothes tight; sudden muffin top
High Libido	Physical	Thinking about sex a lot; desire at its peak
Low Libido	Physical	Lost my drive; sex feels like a chore
Dry Vagina	Physical	Dry down there; parched
Fatigue	Physical	Zombie mode; wiped out; low battery
Hair/Skin Changes	Physical	Going bald; breakouts like teen years
Tremors	Physical	Body vibrating; nervous shakes; cannot hold things steady
Migraines	Cognitive	Splitting headache; throbbing head
Anxiety	Emotional/Psychological	Jittery; on heavy steroids; mind is mush
Mood Swings	Emotional/Psychological	Emotional rollercoaster; snapping at everyone
Depression	Psychological	In the dumps; hopeless; sad for no reason

symptom classification prompts, and model performance was evaluated on a separate set of 250 comments. The term “palpitation” was analyzed and using the keyword method, the model was able to correctly classify 98.3% of cases. However, the F1 score which combines accuracy and recall, was lower at a score of 82.3%. This suggests that although the model is accurate in identifying symptoms, more research is needed to fully evaluate AI’s effectiveness in symptom analysis using this approach.

4 Preliminary Findings

4.1 Informal Language Used to Describe Symptoms

In Table 1, each row represents a specific symptom, categorized into physical, cognitive, emotional, or psychological domain, alongside representative phrases describing how participants describe their symptoms in everyday language. Many described their symptoms using informal phrases like, “blood bath” for heavy flow during periods, “heart jumping” for palpitations, or “dry down there” to describe dry nether regions.

4.2 Identifying Symptoms

Table 1 helped shape a more inclusive symptom list to perform keyword analysis which then gave a list of 67 symptoms (Figure 1). This list indicates a broad range of physical, emotional, cognitive, and psychological symptoms. “Fluctuating periods” is the most frequent symptom which is 3-4x more frequent than the next symptom. Among psychological symptoms, Anxiety, ADHD, anger, panic, depression, stress, brain fog, collectively received 8,500 mentions. Low/high libido, frozen shoulder, tinnitus, vertigo, fatigue and body odor have a smaller but important subset.

4.3 Symptom Patterns

Figure 2 represents symptom clusters. Lower joins in the graph depict that they are often mentioned together, and higher joins indicate that they are less related. The large splits in the dendrogram suggest the presence of a small number of broader symptom domains, rather than all symptoms being closely related. The graph

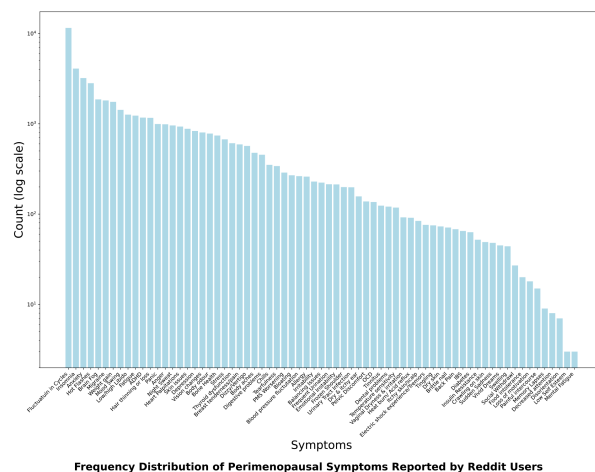


Figure 1: Frequency Distribution of Perimenopausal Symptoms Reported by Reddit Users.

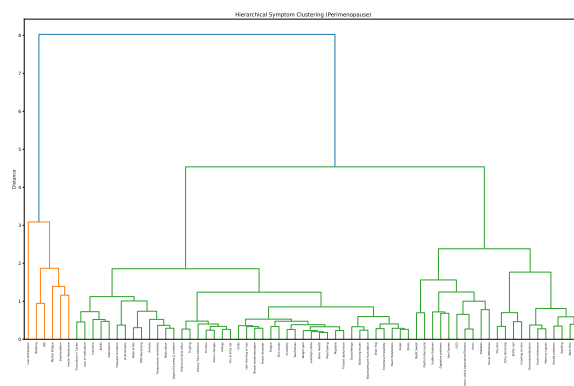


Figure 2: Hierarchical clustering dendrogram showing groups of perimenopausal symptoms that frequently co-occur.

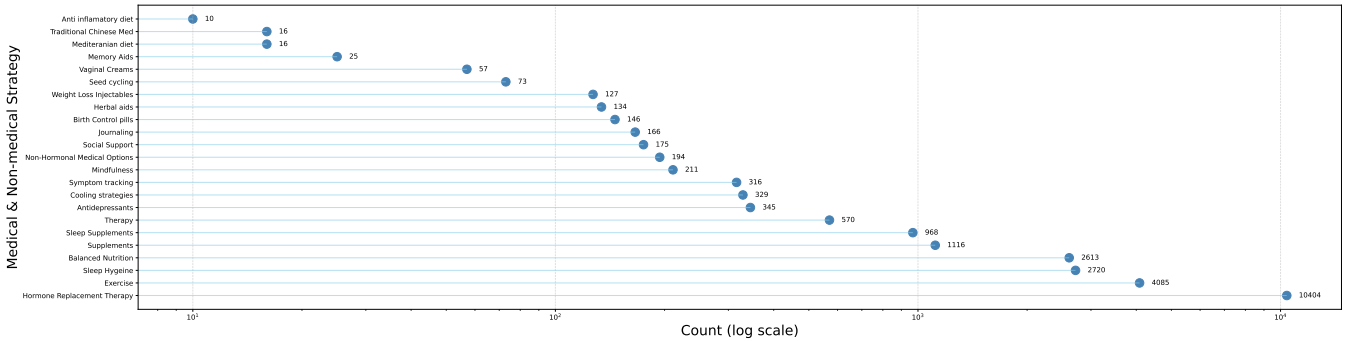


Figure 3: Frequency of medical and non-medical coping strategies reported by women during perimenopause (log scale).

reveals that cognitive and emotional symptoms (e.g. brain fog, anxiety, depression) form a tighter cluster, suggesting a strong association between mental and affective experiences during this phase. Sleep and energy related symptoms cluster closely with fatigue and irritability, highlighting the interplay between rest disruption and emotional well-being. Overall, this graph shows that symptoms are often discussed and experienced in clusters rather than in isolation, which can guide research, screening, or treatment strategies.

4.4 Medical and Non-medical Strategies

Figure 3 illustrates a wide range of strategies that women rely on to navigate this stage, spanning from medical treatments, alternative therapies, and home-based remedies. This analysis found higher levels of discussion around hormone replacement therapy (HRT) and supplements. Others emphasized non-medical approaches such as exercise, eating balanced nutrition, journaling, and mindfulness practices. Sleep related issues were commonly managed with supplements like magnesium and melatonin, as well as herbal aids like ashwagandha and chamomile tea. Cooling methods such as air conditioner, fans, cold showers, and lightweight cotton clothing are commonly used to relieve hot flashes.

5 Information Design

In addition to qualitative graphs, a synthesized infographic (Figure 4) is designed that depicts collective symptomatology. This infographic is the first step towards future work. While the graph (Figure 2) supports detailed exploration and comparison of symptom frequencies, the infographic shows the symptoms collectively from an experiential overview of how symptoms are collectively experienced. The style of visualization is based on the principles of narrative-based visualization and data humanism, where the emphasis is on experiences and meaning instead of abstract representations alone. Together, these visual supports both analytical interpretation and intuitive understanding which makes the data more approachable and relatable which would help women better prepare for this stage.

6 Future Work

Next steps include analyzing ethnicity and lifestyle based phenotypes to gain a better understanding of how perimenopause is experienced differently across diverse populations. In addition, a

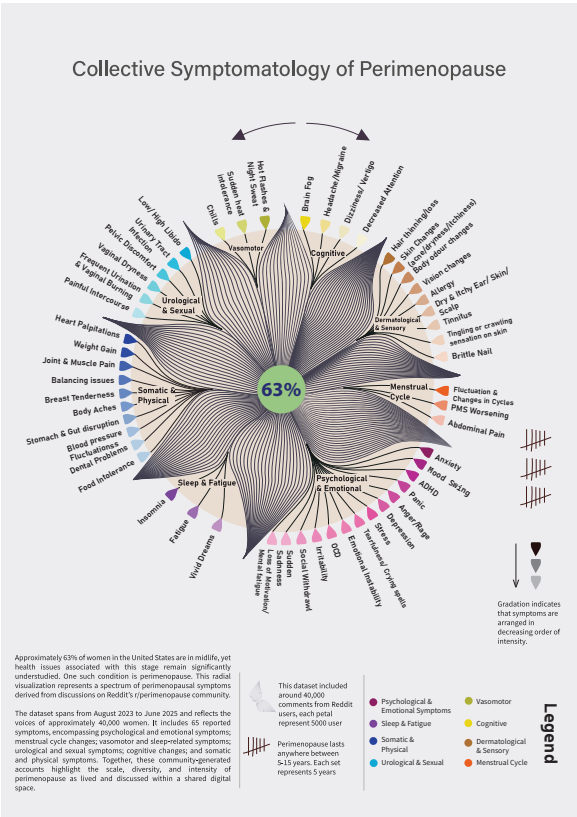


Figure 4: Collective Symptomatology of Perimenopause

workshop will be conducted with women currently in this life stage to gain deeper insight into their varied experiences and explore how they track day to day data- through journaling, drawing, voice notes, apps, or other methods, and if these practices reveal meaningful health patterns. Based on all these findings, a digital repository consisting of interactive data visualizations would be developed to support women navigating this stage, while also serving secondary users like physicians, researchers, and other practitioners, who could also benefit from this repository by getting accessible, data-driven insights into this life stage.

7 Conclusion & Discussion

More broadly, this study provides early steps in the process of validation toward the lived experience-based data visualization to get a holistic understanding of this stage in HCI. Centering women's voices offers valuable insights that can inform the design of healthcare tools and form a deeper understanding of this phase. The goal is to not chew all the data produced in these forums but to understand these experiences and look at patterns which can guide future work. Further, if AI-assisted approaches are used in systematic ways alongside data analysis methodologies, it may offer value in analyzing large-scale, unstructured qualitative data; however, further research is required to fully evaluate the reliability and generalizability of AI-based symptom analysis in this context.

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